



ENTERPRISE AI & HYBRID CLOUD IMPACT

Dharmendra Singh

Cisco Technical Lead · Principal Engineer · 17+ Years of Enterprise AI & Cloud Platform Delivery

Enterprise AI-Driven Platform Leader · Hybrid Cloud Modernization · Scalable Developer Automation

Enterprise engineering leader with 17+ years building mission-critical systems that scale across finance, IoT, and Cisco hybrid cloud. I deliver AI-ready platforms, modular architecture, and automated workflows that reduce cost, accelerate delivery, and improve operational resilience.

17+
YEARS XP

18+
TEAM LED

40%
DEVELOPMENT COST CUT

50%
REVIEW EFFICIENCY

2×
RE:INVENT

✉ **CONTACT**

LINKEDIN ↗

GITHUB ↗

About Me

I'm a **Principal Software Engineer and Technical Team Lead** at Cisco, leading a global team of 14+ engineers across US, India, China, Australia, and UK. With 17+ years in enterprise software, I deliver AI-native engineering, hybrid cloud modernization, and enterprise platform automation that improve operational resilience, reduce cost, and accelerate technology adoption.

I focus on architecture-first systems, modular platforms, and AI-enhanced automation that make complex enterprise infrastructure easier to operate and scale. My work has modernized Cisco cloud deployments, enabled AI-ready applications with minimal code change, and established practices now adopted broadly across the organization.

Outside engineering: I'm a **gardener, father, amateur carpenter, and trekker**. I volunteer with Habitat for Humanity, have judged Cisco-sponsored university hackathons, and cook considerably more than I should admit.



Gardening



Trekking



DIY Carpentry



Cooking



Clean Architecture



Gen AI



UX Design



Habitat for Humanity



LOCATION

Morrisville, North Carolina, USA



EDUCATION

B.Tech — Electrical & Electronics Eng.

VIET, UPTU, India · 2003–2007

· Hindi — Native / Bilingual

 CURRENTLY AT

Technical Team Lead

Cisco Systems · Jun 2020 – Present

 LEADERSHIP FOCUS

Enterprise AI Engineering · Hybrid Cloud Modernization · Scalable Platform Systems

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Technical Arsenal

FRONTEND

React.js

TypeScript

JavaScript

HTML5

CSS3

Sencha ExtJS

AngularJS

Angular 2

Kendo UI

Bootstrap

BACKEND

C# / .NET Core

ASP.NET MVC

Node.js

GoLang

Python FastAPI

REST APIs

Microservices

Java

CLOUD & DEVOPS

AWS

Azure

Docker

Kubernetes

AI & GEN AI

LLMs / Gen AI

OpenAI APIs

Transformer Arch

ARCHITECTURE

- DDD
- TDD
- Clean Architecture
- Design Patterns
- Low-Code Platforms
- SDLC
- Agile / Scrum

IOT & HARDWARE

- Z-Wave
- Zigbee
- Bluetooth
- Alexa for Business
- Edge Computing
- Smart Home

DATA & DATABASES

- SQL Server
- MongoDB
- Oracle
- NoSQL
- Sybase
- Tableau

TOOLS & QUALITY

- Git / GitHub
- Jira / Asana
- VS Code
- Jest
- SonarQube / Scava
- Adobe XD
- Postman
- Webex

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Career Journey



Jun 2020 - Present · 5+ yrs

Cisco Systems

Technical Team Lead · Raleigh, NC (Hybrid)

Lead a global engineering team of 14+ across US, India, China, Australia, and UK, delivering hybrid cloud infrastructure products at Cisco scale within two-week sprint cadences.

- Build proof-of-concept solutions to close strategic gaps, accelerate decision-making, and translate leadership requirements into scalable hybrid cloud architecture
- Designed and launched HAI, a Cisco enterprise AI integrator platform that unified bots, RAG, local LLMs, containerized models, and voice interfaces with minimal application change
- Enabled legacy and non-AI applications to become AI-ready through embeddable script integration, reducing workflow fragmentation and improving internal developer productivity
- Engineered agent orchestration using internal Drift APIs for RAG, BridgeIT for LLM access, SSO token authorization for internal APIs, Splunk observability, Python FastAPI services, Hugging Face local models, and LangChain-based LLM telemetry
- Institutionalized AI harness engineering by creating apm.md guidelines, copilot-instruction and agent.md specifications, reusable skill definitions, and MCP server patterns for operational consistency
- Automated engineering workflows with AI agents for PR review, Sonar remediation, Jira issue creation, and documentation updates, reducing review effort by roughly 50%
- Mentored engineers globally in AI-native development, spec-driven delivery, and modular automation while aligning UX, architecture, and product teams
- Represented Cisco at AWS re:Invent 2019 and 2023 and led the team to deliver a production-ready DocEngine AI support agent in Cisco AI Hackathon 2025
- Received application, monetary, and stock awards for AI automation and platform innovation with measurable business impact

Python

FastAPI

Hugging Face

LangChain

Splunk

Docker

Kubernetes

AWS



Cisco GIA Global Excellence Award 2019



Strategy Champion Coin



Jun 2019 - Jun 2020 · 1 yr

Cisco Systems

Senior Engineer · Raleigh, NC

Built next-generation hybrid cloud infrastructure leading 18 developers across India, US East/West Coast, China, Australia, and UK.

- Introduced modularity, reusability, and config-driven application practices as an organizational first, modernizing Cisco Cloud deployment and management apps
- Led a new cloud deployment application from the ground up using the same architecture principles, delivering a second-generation platform with on-demand scalability
- Delivered production applications that have run for 7+ years and serve thousands of users, reducing feature development time by more than 70%
- Established development, Sonar, security, and build rules to ensure consistent, clean, secure, and supportable code across large engineering teams
- Helped other Cisco IT teams modernize legacy apps using the same system and techniques, enabling wider adoption of configurable, platform-driven delivery
- Interviewed, hired, and mentored developers and managers, with mentees later joining AWS, Nvidia, and other major technology companies
- Presented high-impact demos and architecture reviews to directors, vice presidents, and broad stakeholder audiences

GoLang

Node.js

React

MongoDB

Tableau

Drag & Drop



Jan 2017 - Jun 2019 · 2.5 yrs

K4Connect

Senior Frontend Developer · Raleigh, NC (Startup)

Joined a smart home IoT startup building dashboards for device provisioning, monitoring, state management, and fleet reporting for senior living communities across the US.

- Helped integrate Amazon Alexa for Business at enterprise scale in US senior living communities, first of its kind
- Deep hands-on work with Z-Wave, Zigbee, Bluetooth IoT protocols, hardware programming, edge/core computing
- Met with seniors in communities to understand accessibility fears and usability — shaped the product directly

React

Sencha ExtJS

React Native

Vue

Node.js

AWS

IoT

Alexa SDK



Feb 2013 - Jan 2017 · 4 yrs

Fidelity Investments

Technology Lead · via Infosys · Durham, NC

- Led full modernization of legacy capital markets and 401K platforms — C++/ASP/Mainframe → .NET MVC 4 → Angular 2.0
- Gathered requirements from business users, created stories with BAs, designed mockups, led onshore & offshore delivery
- Improved 401K platform scalability and user experience significantly

C#

.NET MVC 4

Kendo UI

AngularJS → Angular 2

SQL Server

REST APIs



Nov 2007 - Jan 2017 · 9+ yrs

Infosys Technologies

Software Engineer → Senior Engineer → Tech Lead · India / US

Recruited one year before college graduation. Trained at the legendary Infosys Mysore campus. An extraordinary journey: Java → Security Auditor → .NET Tech Lead.

- Conducted security audits for major US banks at Offshore Development Centers; aligned code to ISO standards
- Built custom tools to detect and optimize security standards, saving organizations significant compliance penalties
- Grew from Java developer → Security Auditor (2010) → .NET C# Developer → Senior Engineer → Tech Lead

Java

Security Audits

ISO Standards

ASP.NET

C#

SQL Server

VB Script

Notable Projects

◆ CISCO · AI HACKATHON

2025

DocEngine : AI-Powered Human-in-the-Loop Support Agent

Led ideation, architecture, and full development in a Cisco-wide 48-hour hackathon. Built an intelligent agent system combining OpenAI LLMs, Hugging Face packages, and Cisco's ecosystem (Webex, NLP APIs, Vector DBs) to modernize internal support and knowledge retention. Humans guide AI learning via thumbs-up/down feedback — a perpetual human-machine learning loop. Built Webex Bot front-end, full REST API, and modular intent-routing system. Delivered polished product video, live demo, and executive-style presentation for judges.

Python FastAPI

OpenAI APIs

Hugging Face

AI Agents

Agentic Architecture

Vector DB

Webex Bot

NLP

◆ CISCO · 2019-2024

Hybrid Cloud Low-Code Platform

Led development of a next-gen low-code infrastructure platform reducing development effort by 40% and cost by 30%. Designed for maximum reusability, modularity, and decoupling at global Cisco scale. Used by teams globally to ship cloud-native features faster.

React

TypeScript

GoLang

Node.js

Kubernetes

Docker

AWS

◆ K4CONNECT · 2017-2019

Enterprise Alexa for Business – Senior Living IoT

React

Sencha ExtJS

Alexa SDK

IoT

Z-Wave

Node.js

♦ AWS RE:INVENT · 2019 & 2023

Cisco Delegate - AWS re:Invent Las Vegas

Represented Cisco at AWS re:Invent in 2019 and 2023. Explored Gen AI, Bedrock, foundation models, and enterprise cloud strategy, bringing insights back to shape Cisco's AI roadmap.

AWS

Gen AI

Bedrock

Cloud Strategy

♦ FIDELITY · 2013-2017

401K Platform Modernization

Full modernization from C++/Classic ASP/Mainframe → .NET MVC 4 → Angular 2.0 at Fidelity's Durham campus. Designed mockups, led onshore & offshore teams end-to-end.

C#

.NET MVC

Angular 2

Kendo UI

SQL Server

♦ INFOSYS · 2008-2011

Bank Security Audit Tools

Built custom tools to detect, test, and optimize security standards for offshore development centers of major US banks, aligning to ISO standards and saving organizations significant compliance penalties.

Java

Security Audits

ISO Standards

ASP.NET

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What People Say



Dharmendra is an amazon software engineer. A talented frontend developer who creates engaging, user-friendly applications using React. Excellent problem-solving, communication, and collaboration. He fully contributes and is reliable in meeting deadlines and delivering high-quality work.

GD**Guoyang Di**

Software Engineer · Cisco · Jul 2023



Naturally curious, a hard worker who puts in time to understand both the technology and the business. Worked at a large financial firm with deep interdependencies and complex architecture. Communicates very well across a matrix of job roles. I would recommend him without hesitation.

DM**Daniel Mullaney**Principal Systems Analyst · Fidelity
· May 2019

Dharma has so much heart and gives 110% to everything he does. Hard worker who cares deeply about the product and its members. Willing to share thoughts with senior management and always such a positive spirit in the team.



His patience towards new ideas and ability to convert them into action at the lowest level is commendable. His zeal and hunger for betterment is what makes him complete. I strongly recommend him as one of the best team leads.



Dharm was a pleasure to work with. Genuinely optimistic, eager to help others, intelligent but not arrogant. Hard-working, dedicated, knowledgeable, and approachable. He would be an asset to any team.

JP

Jamie Pittman

Senior Software Engineer · K4Connect
· Jun 2019



Energetic, fast, adventurous — a commando-like approach to solving problems. Deep knowledge and a natural ability to impress with expertise in a pleasant, comprehensible manner. I strongly recommend DS for any challenge.

NU

Navaneet Upadhyay

Cybersecurity Lead · Microsoft
Ecosystem Expert · Apr 2020

Certifications & Credentials

**AWS Partner: Technical Accredited**

Amazon Web Services · Dec 2024

**Azure AI-900 AI Fundamentals**

Microsoft · Feb 2024

**Generative AI — Blue Belt**

Cisco Data Science · Mar 2024

**AI/ML Fundamentals**

Cisco · Feb 2024



Generative AI Fundamentals

Google · Dec 2023



Introduction to Large Language Models

Google · Dec 2023



Responsible AI with Google Cloud

Google · Jan 2024



Working with Large Language Models

LinkedIn Learning · Dec 2023



Data Literacy — Green Belt

Cisco · Jul 2022



Agile Fundamentals (ONEx)

Cisco · Apr 2022



Programming in HTML5 with JavaScript & CSS3

Microsoft · Dec 2012



Querying Microsoft SQL Server 2012

Microsoft · May 2013

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Giving Back



HABITAT FOR HUMANITY · JUN 2025

Construction Associate, Raleigh,
NC



NC A&T UNIVERSITY · OCT 2023

Hackathon Organizer & Judge :
HackNCAT 7.0

ocean of an incredible cause for families in need.

student teams, and selected winners. Represented Cisco to inspire the next generation.



HABITAT FOR HUMANITY · APR 2023

Construction Volunteer : Old Poole Place, Raleigh

Assisted with framing, siding, and window installation at the Old Poole Place site. Worked alongside builders and community volunteers to create safe, affordable homes.



HABITAT RESTORE · FEB 2023

Store Volunteer

Restocked and organized Habitat ReStore locations, helped customers, and evaluated incoming merchandise. Every item resold keeps goods from landfills and makes quality things affordable for those who need them.

Let's build something **remarkable** together.

Built enterprise platform systems that improve operational resilience, accelerate delivery, and enable AI-ready infrastructure across global teams.

✉ CALLDHARM@GMAIL.COM

[LINKEDIN](#) ↗

[GITHUB](#) ↗

📍 Morrisville, NC, USA

© 2026 Dharmendra Singh
Building the Future of AI-Driven Engineering

Dharmendra Singh

AI-Native Technical Lead • Enterprise Architect • UI/UX Strategist • React • TypeScript • Cloud-Native • Agentic AI • AWS • Azure • Developer Experience

Technical Lead at Cisco with 17+ years of experience building scalable enterprise platforms, AI-powered developer workflows, cloud-native applications, and modern UI/UX systems for global teams.

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